8.2 Long-term Recurrent Convolutional Networks (LRCNs)

- ideally, an image captioning model should provide variable length outputs, including generation of full-length sentence descriptions that go beyond conventional one-versus-all prediction tasks
- Long-term Recurrent Convolutional Networks (LRCNs) represent a class of architectures for visual recognition and description which combine convolutional layers and long-range temporal recursion and are end-to-end trainable (Figure 3)
- it can be instantiated for specific video activity recognition, image caption generation, and video description tasks
- LRCNs show that convolutional networks with recurrent units are generally applicable to visual time-series modeling
- in visual tasks, LSTM-style RNNs can provide significant improvement when ample training data are available to learn or refine the representation
- specifically, LSTM-type models enable a novel end-to-end optimizable mapping from image pixels to sentence-level natural language descriptions

8.2 Long-term Recurrent Convolutional Networks (LRCNs)

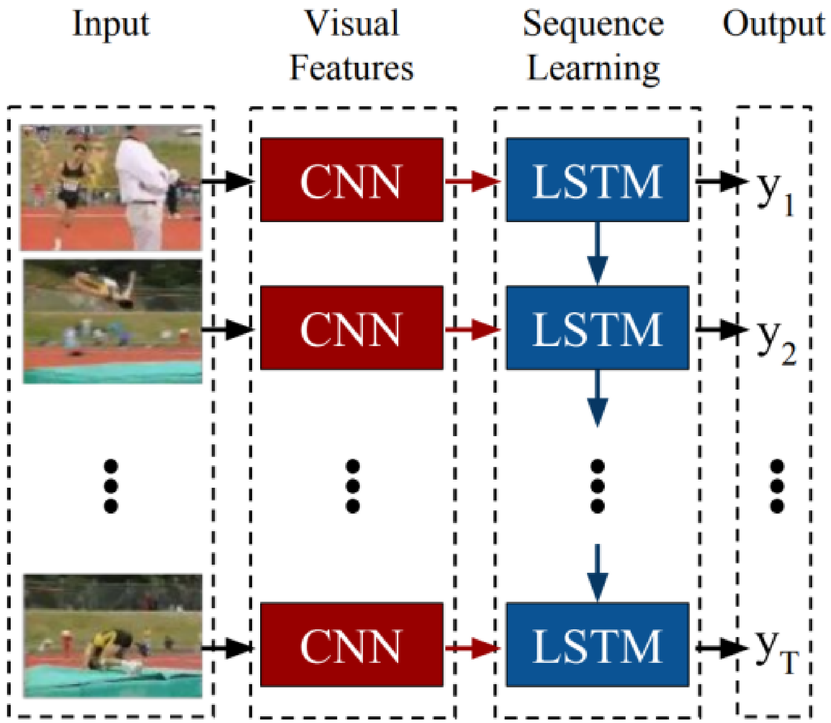


Figure 3: Long-term Recurrent Convolutional Networks (LRCNs), a class of architectures leveraging the strengths of rapid progress in CNNs for visual recognition problems, and the growing desire to apply such models to time-varying inputs and outputs. LRCN processes the (possibly) variable-length visual input (left) with a CNN (middle-left), whose outputs are fed into a stack of recurrent sequence models (LSTMs, middle-right), which finally produce a variable-length prediction (right). Both the CNN and LSTM weights are shared across time, resulting in a representation that scales to arbitrarily long sequences.

8.2 Long-term Recurrent Convolutional Networks (LRCNs)

- LRCNs construct end-to-end trainable image to sentence mappings
- machine translation models are encoder-decoder pairs based on LSTM networks
- LRCNs are a multimodal analog of this model, and their architecture uses a visual CNN to encode a deep state vector, and an LSTM to decode the vector into a natural language string (Figure 4)
- the resulting model can be trained end-to-end on large-scale image and text datasets

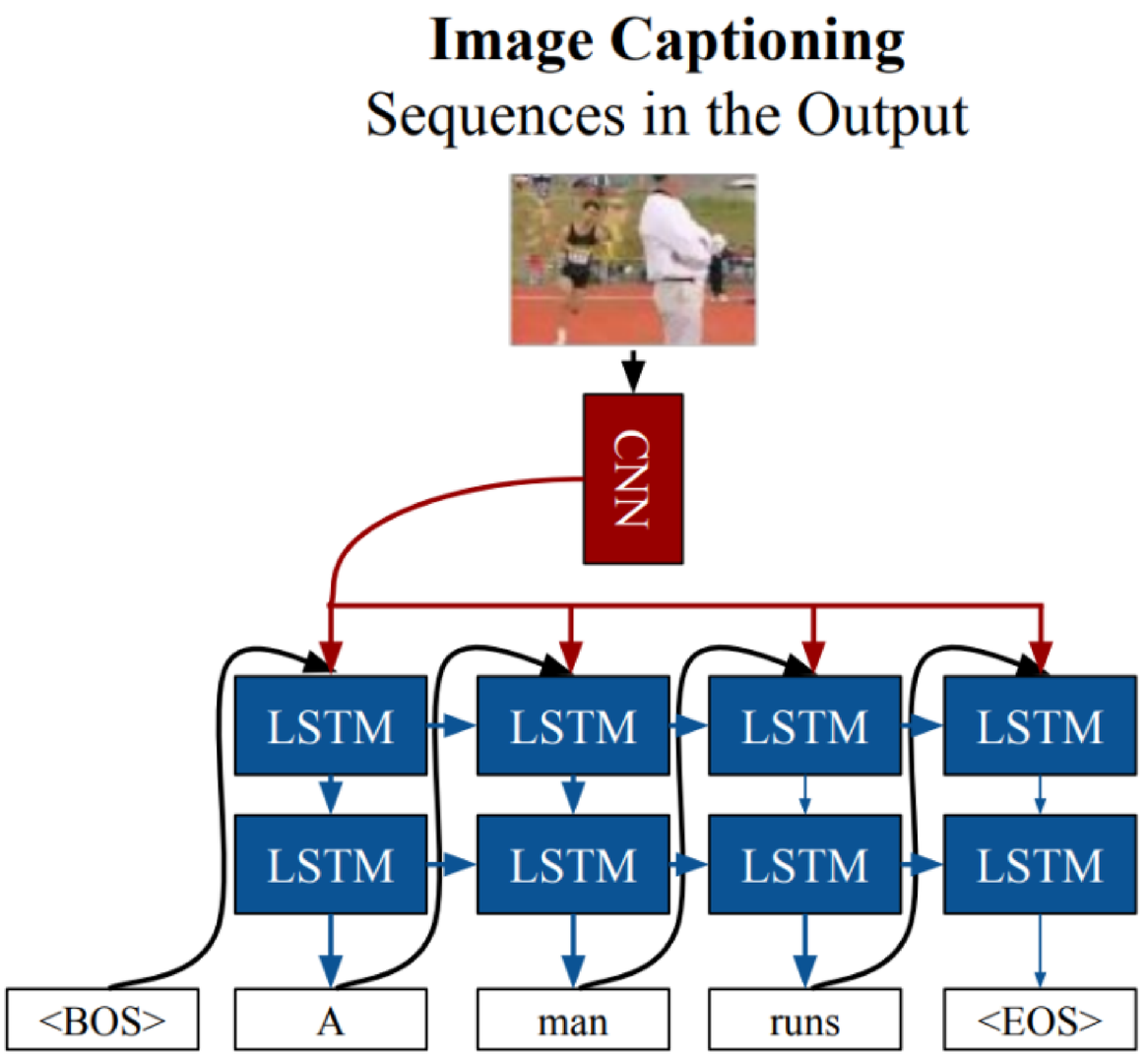


Figure 4: Task-specific instantiation of the LRCN model for image captioning.

8.2 Long-term Recurrent Convolutional Networks (LRCNs)

- LRCN works by passing the visual input x (an image) through a feature transformation $\phi_V(\cdot)$ with parameters V, usually the activations in some layer of a deep CNN, to produce a fixed-length vector representation $\phi_V(x)$
- the outputs of ϕ_V are then passed into a recurrent sequence learning module
- some models stack multiple LSTMs on top of each other, as described later
- to predict a distribution $P(y_t)$ over outcomes $y_t \in C$ (where C is a discrete, finite set of outcomes) at time step t, the outputs of the sequential model are passed through a linear prediction layer
- finally, the predicted distribution $P(y_t)$ is computed by taking the softmax of the linear prediction layer
- the image captioning problem represents the mapping $x \mapsto \langle y_1, y_2, \ldots, y_T \rangle$, with a static image as input, but a much larger and richer label space consisting of sentences of any length

8.2 Long-term Recurrent Convolutional Networks (LRCNs)

- in order to perform image captioning, at each time step, both the image features and the previous word are provided as inputs to the sequence model, in this case a stack of LSTMs (each with 1000 hidden units), which is used to learn the dynamics of the time-varying output natural language sequence
- at time step t, the input to the bottom-most LSTM in Figure 4 is the embedded word from the previous time step y_{t-1}
- input words are encoded as "one-hot" vectors: vectors $y \in \mathbb{R}^K$ with a single non-zero component $y_i = 1$ denoting the ith word in the vocabulary, where K is the number of words in the vocabulary, plus one additional entry for the <BOS> (beginning of sequence) token, which is always taken as y_0, the "previous word" at the first time step ($t = 1$)
- these one-hot vectors are then projected into an embedding space with dimension d_e by multiplication $W_e y_t$ with a learned parameter matrix $W_e \in \mathbb{R}^{d_e \times K}$
- the result of a matrix-vector multiplication with a one-hot vector is the column of the matrix corresponding to the index of the single nonzero component of the one-hot vector
- W_e can therefore be thought of as a "lookup table", mapping each of the K words in the vocabulary to a d_e-dimensional vector

8.2 Long-term Recurrent Convolutional Networks (LRCNs)

- the visual feature representation $\phi_V(x)$ of the image x may be input to the sequence model – a stack of L LSTMs – by concatenating it at each time step either with (1) the embedded previous word $W_e y_{t-1}$ and fed into the first LSTM of the stack, or (2) the hidden state $h_t^{(\ell-1)}$ output from LSTM $\ell - 1$ and fed into LSTM ℓ, for some $\ell \in \{2, \ldots, L\}$
- these choices are depicted in Figure 5
- we refer to the latter choice as "factored", as it forces a sort of separation of responsibilities by "blinding" the first $\ell - 1$ LSTMs and forcing all of the capacity of their hidden states at time step t to represent only the partial caption $y_{1:t-1}$ independent of the visual input, while the LSTMs starting from ℓ are responsible for fusing the lower layer's hidden state given by the partial caption with the visual feature representation $\phi_V(x)$ to produce a joint hidden state representation $h_t^{(\ell)}$ of the visual and language inputs up to time step t from which the next word y_t can be predicted
- in the factored case, the hidden state h_t for the lower layers is conditionally independent of the image x given the partial caption $y_{1:t-1}$

8.2 Long-term Recurrent Convolutional Networks (LRCNs)

- the outputs of the final LSTM in the stack are the inputs to a learned linear prediction layer with a softmax producing a distribution $P(y_t|y_{1:t-1}, \phi_V(x))$ over words y_t in the model's vocabulary, including the <EOS> token denoting the end of the caption, allowing the model to predict captions of varying length
- the visual model ϕ_V used is the VGG model pre-trained for ImageNet classification
- without any explicit language modeling or impositions on the structure of the generated captions, the described LRCN system learns mappings from images input as pixel intensity values to natural language descriptions that are often semantically descriptive and grammatically correct
- at training time, the previous word inputs $y_{1:t-1}$ at time step t are from the ground truth caption
- for inference of captions on a novel image x, the input is a sample $y_t \sim P(y_t|y_{1:t-1}, \phi_V(x))$ from the model's predicted distribution at the previous time step, and generation continues until an <EOS> (end of sequence) token is generated

8.2 Long-term Recurrent Convolutional Networks (LRCNs)

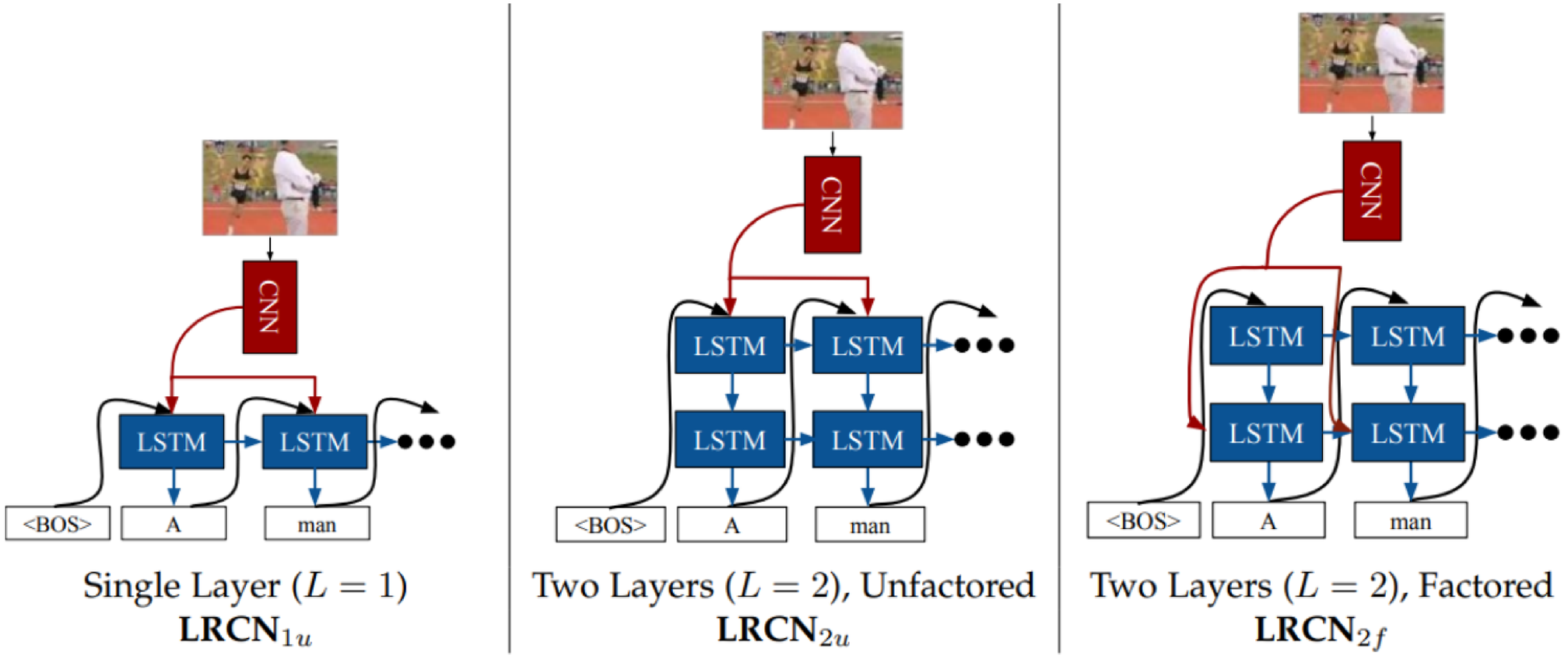


Figure 5: Three variants of the LRCN image captioning architecture. It is explored the effect of depth in the LSTM stack, and the effect of the "factorization" of the modalities.